UNIT II

Spur Gear Design

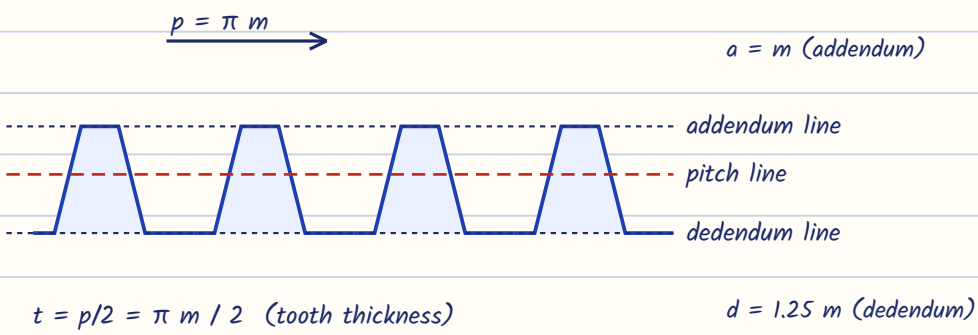
gear geometry | kinematics | forces on tooth | stresses | material selection (bending & contact) | design | power capacity

I. Introduction to gears

Gears are toothed wheels that transmit power and motion between shafts with a **constant velocity ratio** and no slip. In an automobile, gears are used in the **gearbox, differential, final drive, timing drive and steering**. **Spur gears** have teeth cut **parallel to the shaft axis**; they connect **parallel shafts**; simplest and cheapest, but noisy at high speed.

Classification	Types
By shaft position	Parallel: spur, helical, herringbone, rack & pinion. Intersecting: bevel. Non-parallel non-intersecting: worm, crossed helical, hypoid.
By peripheral velocity	Low (< 3 m/s), medium (3-15 m/s), high (> 15 m/s).
By meshing	External, internal (annular), rack and pinion.
By tooth profile	Involute (standard), cycloidal.

II. Gear geometry



Term	Meaning / relation
Pitch circle (d)	Imaginary circle that rolls without slip with the mating circle. $d = m\,z$
Module (m)	$m = d / z$ (mm); the basic size of the tooth. Standard (IS 2535): 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12, 16, 20 …
Circular pitch (p)	Arc distance between corresponding points of adjacent teeth on pitch circle: $p = \pi d / z = \pi m$
Diametral pitch (P)	$P = z / d = 1 / m$ (teeth per inch in inch system)
Addendum (ha) / **Dedendum** (hf)	Radial heights above / below pitch circle: $ha = m$; $hf = 1.25\,m$ (for 20° full depth)
Clearance (c)	$c = hf - ha = 0.25\,m$
Tooth thickness / space	$t = p/2 = \pi m / 2$ (theoretical); backlash = space width - mating tooth thickness
Pressure angle (φ)	Angle between the common normal (line of action) and the common tangent at the pitch point; standard 20° (14.5° older)
Base circle (db)	$db = d \cos \varphi$: circle from which the involute is generated; base pitch $pb = \pi m \cos \varphi$
Outside / root diameter	$da = m (z + 2)$; $df = m (z - 2.5)$
Centre distance (a)	$a = (d1 + d2)/2 = m (z1 + z2)/2$
Face width (b)	Width of tooth along the axis: $b = 9.5\,m$ to $12.5\,m$ ($\approx 10\,m$ typical); $b \le d1$ (pinion)
Gear ratio (i)	$i = z2/z1 = N1/N2 = d2/d1 \ge 1$ (reduction); up to 6-8 per stage for spur gears

Standard tooth systems

System	φ	Addendum	Dedendum	Remark
14.5° composite / full depth	14.5°	m	1.25 m	Old; min 32 teeth to avoid interference
20° full depth	20°	m	1.25 m	Most used; stronger tooth, min 17 teeth
20° stub	20°	0.8 m	1.0 m	Very strong, short; min 14 teeth; less sliding

III. Law of gearing and involute profile

> **Law of gearing**: for a constant velocity ratio, the **common normal at the point of contact must always pass through the fixed pitch point** P on the line of centres.

> **Involute** = locus of a point on a taut string unwrapped from a base circle. It satisfies the law; the line of action is the straight line tangent to both base circles (inclined at φ).

> Advantages: **centre distance can vary slightly without affecting velocity ratio**, easy to manufacture by hobbing, constant pressure angle.

> Involute function: inv φ = tan φ − φ (radians). Involute pressure angle at radius r: cos φ_r = rb/r.

Kinematics of meshing

Quantity	Formula
Velocity ratio	N1/N2 = ω1/ω2 = d2/d1 = z2/z1
Pitch line velocity	v = π d1 N1 / 60000 (m/s, d in mm, N in rpm)
Path of contact	Approach + recess: = √(Ra1² − Rb1²) + √(Ra2² − Rb2²) − a sin φ
Contact ratio (CR)	CR = (length of path of contact) / (base pitch pb) ; for spur gears ≥ 1.2 (1.4–1.6 good). It = average number of teeth in contact.
Sliding velocity	Zero at pitch point; max at start & end of contact; Vs = (ω1 + ω2) × (distance from pitch point)

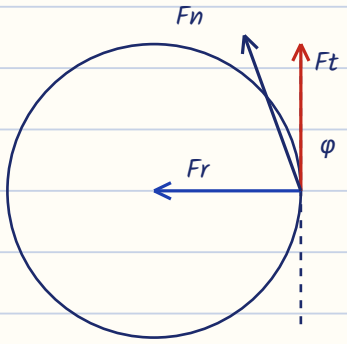
Interference & undercutting

> **Interference**: the tip of one tooth cuts the root of the other because contact occurs beyond the tangent points (base circles). It causes undercutting -> weakens the tooth.

> Minimum teeth to avoid interference on a pinion meshing with a rack: z_min = 2 ha*/sin²φ = **17 (20° FD), 14 (20° stub), 32 (14.5°)**. Smaller number is possible against a larger gear (about 13 for equal 20° gears).

> Remedies: stub teeth, larger pressure angle, **profile shift (addendum modification)**, undercut the pinion root on purpose, increase the pinion teeth.

IV. Forces on a gear tooth



Ft = 2T / d = 2000 P / (π d N/60)

Fr = Ft tan φ

Fn = Ft / cos φ

Ft transmits power; Fr only loads shaft & bearings (no work done).

φ = pressure angle (20°)

Force	Formula (spur gear)
Torque on pinion	$T = 60\,P / (2\pi N) = 9.55 \times 10^6\,P / N$ (N·mm; P in kW, N in rpm)
Tangential force Ft	$Ft = 2\,T / dl = 2\,T / (m\,zl) = 1000\,P / v$ (v in m/s)
Radial force Fr	$Fr = Ft \tan \varphi$
Normal (total) force Fn	$Fn = Ft / \cos \varphi = \sqrt{(Ft^2 + Fr^2)}$
Load on shaft/bearing	Resultant Fn; shaft is bent by Fr (in plane) and Ft (perpendicular plane) and twisted by T.

> On the driven gear the forces are equal and opposite; if the pinion is overhung, the shaft bending increases.

> Using direction: driving gear force opposes rotation at tooth; shaft reactions by taking moments in the plane of Ft and in the plane of Fr.

Example 1 – forces

> P = 7.5 kW at 1000 rpm; zl = 20, m = 3 mm, φ = 20°. $T = 9.55 \times 10^6 \times 7.5/1000 = 71{,}625$ N·mm.

> dl = 60 mm -> Ft = 2T/dl = 2387 N ; Fr = 2387 tan 20° = 869 N ; Fn = 2387/cos 20° = **2540 N**.

V. Stresses in gear tooth

1. Bending stress – Lewis equation

Wilfred Lewis (1892) treated the tooth as a **cantilever beam** loaded by Ft at the tip, with a parabola (weakest section) of thickness t at height h: $\sigma b = M/Z = (Ft\ h) / (b\ t^2/6) = 6\ Ft\ h / (b\ t^2)$. Writing $t^2 = 4 \times h$ ... gives:

$$\sigma b = Ft / (b\ m\ Y) \quad \text{with} \quad Y = \pi\ y \ \text{(Lewis form factor)}$$

z	15	17	20	25	30	40	60	100	rack
y (20° FD)	0.093	0.100	0.108	0.117	0.124	0.131	0.139	0.145	0.154

> Formulae: $y = 0.154 - 0.912/z$ (20° FD); $y = 0.124 - 0.684/z$ (14.5° composite); $y = 0.175 - 0.841/z$ (20° stub).

> **Beam strength** $Sb = \sigma b\ m\ b\ Y$ (N) – the load the tooth can carry without breaking. The weaker of pinion / gear (smaller $\sigma b \times Y$) governs: compare $\sigma b1\ y1$ and $\sigma b2\ y2$.

> Valid for the load at the tip – conservative; actual contact at pitch point of two teeth pairs (CR > 1).

2. Dynamic load (velocity factor)

Errors in profile and pitch, elasticity and inertia produce **impact**, so the actual load is higher than Ft.

Barth's equation uses a velocity factor Cv:

Gear quality	Cv (v in m/s)	Range
Ordinary / cut gears	$Cv = 3 / (3 + v)$	v < 10 m/s
Carefully cut, hobbed	$Cv = 4.5 / (4.5 + v)$	v up to 20 m/s
Accurately hobbed & generated	$Cv = 6 / (6 + v)$	v up to 20 m/s
Precision ground / shaved	$Cv = 5.6 / (5.6 + \sqrt{v})$	v > 20 m/s

> **Effective (dynamic) load** $Fd = Ft / Cv$ (or $Cs\ Ft / Cv$ with service factor Cs: 1.0 uniform, 1.25 light shock, 1.5 medium, 1.75–2.0 heavy).

> **Buckingham**: $Fd = Ft + Fi$; $Fi = 21\ v\ (C\ b + Ft) / (21\ v + \sqrt{(C\ b + Ft)})$, with C = deformation factor ≈ 11,860 e for steel pair (20° FD), e = tooth error (mm).

> Design rule: **Sb ≥ Fd** (bending) and **Sw ≥ Fd** (wear).

3. Surface (contact) stress – wear

Pitting (surface fatigue) is caused by Hertz contact stress between two cylinders of radii ρ1 and ρ2 at the pitch point ($\rho = (d/2) \sin \varphi$):

$$\sigma c = \sqrt{[\ (Fn / \pi\ b)\ (1/\rho 1 + 1/\rho 2)\ /\ (\ (1 - \nu 1^2)/E1 + (1 - \nu 2^2)/E2\ )\ ]} \leq \sigma c,allow$$

> For steel-steel ($E = 2.06 \times 10^5$ MPa, $\nu = 0.3$): $\sigma c \approx 190 \sqrt{[\ Fn\ (1/\rho 1 + 1/\rho 2)\ /\ b\ ]}$ (MPa; N, mm).

> **Buckingham wear load** $Sw = b\ Q\ d1\ K$ where $Q = 2\ z2/(z1 + z2) = 2\ i/(i + 1)$ (ratio factor; external gears) and $K = (\sigma es)^2 \sin\varphi\ (1/E1 + 1/E2)/1.4$ = load-stress factor.

> Approximately $K = 0.16\ (BHN/100)^2$ N/mm² for steel on steel (BHN = average hardness of the pair, or softer member).

> σes = surface endurance limit ≈ (2.8 BHN - 70) MPa for steel (take other materials from the data book).

> Condition: **Sw ≥ Fd**.

VI. Selection of gear material

Material must resist **bending fatigue at root** (strength, toughness) and **pitting wear at flank** (surface hardness), be machinable, damping, cheap. Choose according to **σb (bending stress)** and **σc / BHN (contact stress)**.

Material	σut (MPa)	BHN	Allowable σb ≈ σut/3 (MPa)	Use
Grey cast iron FG200	200	180	67	Low-cost, slow, quiet; large gears
Grey cast iron FG260	260	200	87	Medium duty, good damping
Steel C30 / C40 (normalised)	500-580	150-180	170-190	General, light-medium loads
Steel C45 / 40C8 (hardened & tempered)	600-750	200-250	200-250	Medium - heavy; most common
Alloy steel 40Cr1 / 40Ni3	800-1000	250-300	270-330	Heavy duty, shock
Case-hardened 20MnCr5 / 16MnCr5 / 17Cr	900-1100 (core)	surface 55-62 HRC	(case) 300-400	**Automobile gears**, high power density
Nitrided steel (31CrMoV9)	1000+	surface 60 HRC	300-400	Minimum distortion, high precision
Phosphor bronze, Nylon, phenolic (textolite)	low	-	low	Worm wheels; quiet, lightly loaded; timing gears

> Values are typical; use the exact allowable stresses from the PSG / Bhandari design data book for the problem.

> **Pinion harder than gear** by 30-50 BHN (the pinion has more cycles). For same material -> pinion is weaker in bending only if σb1 y1 < σb2 y2: check **σ y** for both.

> **Selection logic**: (1) estimate Ft; (2) compare Sb and Sw requirement; (3) if wear governs raise hardness (heat treatment, case hardening); if bending governs raise σut or module/face width.

> Heat treatments: through-hardening (BHN ≤ 350), flame/induction hardening, carburising + quench (58-62 HRC), nitriding.

VII. Design of spur gears

Given: power P, pinion speed N1, gear ratio i, working conditions (shock, hours). **Required**: module m, teeth z1 z2, face width b, material, checks.

1. Choose **materials** and heat treatment -> σb1, σb2, BHN. Choose pressure angle 20° FD, $z1 \geq 17$ (usually 18-24), $z2 = i\ z1$ (round).
2. Design torque $T = 9.55 \times 10^6\ P\ Cs / N1$ (N·mm) with service factor Cs.
3. Assume face width $b = 10\ m$ ($b/m = 9.5$-12.5). Pitch line velocity $v = \pi\ m\ z1\ N1/60000$ -> Cv.
4. Find y1, y2 -> $Y = \pi\ y$; weaker gear = smaller (σb × y). Design the **weaker** one for **beam strength**: $Ft = \sigma b\ Cv\ b\ \pi\ y\ m$.
5. Solve for module: from $Ft = 2T/(m\ z1)$ and $Ft = \sigma b\ Cv\ (b/m)\ \pi\ y\ m^2$:

$$m^3 = 2\ T / [\ z1\ \sigma b\ Cv\ (b/m)\ \pi\ y\ ] \quad \text{-> take next standard module}$$

6. Fix $b = 10\ m$, $d1 = m\ z1$, $d2 = m\ z2$, $a = (d1 + d2)/2$, recompute v, Cv, $Fd = Ft/Cv$.
7. **Check wear**: $Sw = b\ Q\ d1\ K \geq Fd$; if not, increase hardness / b / m or change material.
8. Check bending: $Sb = \sigma b\ m\ b\ Y \geq Fd$. Also check **contact ratio, gear blank dimensions, shaft & key**.
9. Specify: module, z, face width, material & heat treatment, grade of accuracy, lubrication.

Example 2 – complete design

> Design a pair of spur gears to transmit 7.5 kW at 1000 rpm, $i = 3$, 20° full depth. σb (both) = 160 MPa; $z1 = 20$, $z2 = 60$, $Cs = 1$, ordinary cut gears ($Cv = 3/(3 + v)$).

> $T = 9.55 \times 10^6 \times 7.5/1000 = 71{,}625$ N·mm ; $y = 0.154 - 0.912/20 = 0.1084$; $Y = \pi y = 0.3405$; $b = 10\ m$.

> $m^3\ Cv = 2T/[z1\ \sigma b\ (b/m)\ \pi\ y] = 143{,}250/[20 \times 160 \times 10 \times 0.1084 \times \pi] = 13.15$. Pitch velocity $v = 1.047\ m$ (m/s). Try $m = 3$: $v = 3.14$ m/s, $Cv = 3/6.14 = 0.489$, $m^3\ Cv = 27 \times 0.489 = 13.2$ ✓ -> **$m = 3$ mm**.

> $d1 = 60$, $d2 = 180$, $a = 120$ mm, $b = 30$ mm. $Ft = 2T/d1 = 2388$ N ; $Fd = Ft/Cv = 4886$ N. $Sb = 160 \times 3 \times 30 \times 0.3405 = 4903\ N \geq Fd$ ✓.

> Wear: $Q = 2 \times 3/4 = 1.5$; for BHN = 350, $K = 0.16\ (3.5)^2 = 1.96$; $Sw = b\ Q\ d1\ K = 30 \times 1.5 \times 60 \times 1.96 = 5292\ N \geq 4886$ ✓ -> pinion & gear hardness about 350 BHN.

VIII. Power transmitting capacity

Power capacity of a given pair = the maximum power it can transmit safely. It is limited by the **lower** of the beam strength and wear strength.

$$P = Ft_allow \cdot v / 1000 \ \ (kW); \quad Ft_allow = Cv \cdot \min(Sb, Sw) / Cs$$

> Use the **weaker member** (pinion or gear) for each failure mode: $Sb = \sigma b\ m\ b\ Y$; $Sw = b\ Q\ d1\ K$.

> Power $\propto v$, so power capacity rises with speed up to the point where dynamic load (Cv) cuts it: capacity is maximum near $v = 10$–15 m/s for ordinary gears.

> Increase capacity by: harder surface, larger module/face width ($b \leq 14\ m$ to avoid misalignment), better accuracy (higher Cv), better lubrication, helical gears.

Example 3 – power capacity

> Using the gear pair of Example 2: $Sb = 4903$ N, $Sw = 5292$ N -> min = 4903 N. $Ft_allow = Cv \times 4903 = 0.489 \times 4903 = 2398$ N.

> $P = Ft_allow \times v / 1000 = 2398 \times 3.14 / 1000 =$ **7.53 kW** ≈ design power ✓.

IX. Gear tooth failures

Failure	Cause	Remedy
Tooth breakage	Overload / bending fatigue at root; shock; misalignment	Larger module / face width, tougher steel, fillet radius, accurate mounting
Pitting	Contact (Hertz) fatigue of the surface near pitch line	Higher hardness, better lubricant, larger gear dia, improved finish
Scoring / scuffing	Breakdown of oil film at high speed & load; welding of asperities	EP additives, lower temp, surface treatment, smoother profile
Abrasive wear	Dirt / foreign particles in the lubricant	Filtration, seals, clean oil
Plastic flow / rolling	Overload on soft material (cast iron / soft steel) flows the surface	Harder material, reduce load
Corrosive wear	Acids / moisture	Correct lubricant, sealing, coatings

Lubrication and general design rules

> **Splash** lubrication for v < 12 m/s (oil level 1-3 teeth deep); **pressure spray / jet** for higher speeds. Oil: SAE 80W-90 / EP gear oil (gearbox).

> Gear wheel construction: solid disc (d < 200 mm), web with holes (200-500 mm), arms (> 500 mm, elliptical 4-6 arms, cast).

> **Backlash** (clearance between non-working flanks) ≈ 0.04 m - 0.1 m for smooth running and thermal expansion.

> Accuracy grade (IS 3756): grade 6-8 for automobile gears; hobbing, shaping, shaving, grinding to improve.

> Face width limit: b ≥ 9.5 m (good contact) and b ≤ 12.5 m typical. Pinion with **z1 ≥ 17** to avoid undercut.

IX-A. Additional topics & solved examples

Contact ratio - worked example

z1 = 20, z2 = 60, m = 3, φ = 20°. r1 = 30, r2 = 90, a = 120 mm. Ra1 = 33, Ra2 = 93 ; Rb1 = 30 cos 20° = 28.19, Rb2 = 84.57. Path of contact = $\sqrt{(33^2 - 28.19^2)} + \sqrt{(93^2 - 84.57^2)} - 120 \sin 20° = 17.15 + 38.69 - 41.04 = 14.80$ mm. Base pitch $p_b = \pi m \cos \varphi = 8.857$ mm. **CR = 14.80/8.857 = 1.67** (> 1.2 ✓ - 1 to 2 pairs in contact).

Which gear is weaker? - comparison of σ y

Pinion C45 (σb = 200 MPa) z1 = 18: y1 = 0.154 - 0.912/18 = 0.1033 -> σb1 y1 = 20.7. Gear grey CI FG260 (σb = 87 MPa) z2 = 54: y2 = 0.154 - 0.912/54 = 0.1371 -> σb2 y2 = 11.9. Smaller product = **gear is weaker**, so the module is designed for the cast-iron gear (use its σb and y in Ft = σb Cv b π y m); then check the pinion.

Profile shift (addendum modification)

- Move the cutter away from the blank by x m: **positive shift** thickens the root of a small pinion and avoids undercut.
- Minimum shift to avoid undercut: x_min = (17 - z)/17 (20° FD). Eg z = 12: x = 0.29 m.
- Centre distance changes with shifts; sum of shifts (x1 + x2) = 0 keeps a = standard (equal & opposite shifts: 'S0' gears).

Rack & pinion, internal gears

Gear type	Notes
Rack and pinion	Rack = gear of infinite radius; straight teeth. Linear travel = π d N; steering gear.
Internal gear	Teeth on inside of a ring; centre distance a = m (z2 - z1)/2; both rotate in same direction; compact, strong (Q = 2 z2/(z2 - z1)); used in planetary gear trains.
Idler gear	Does not change ratio, only direction: reverse gear in automobile gearbox.

Manufacture of gears

Method	Process	Quality
Casting	Sand casting, die casting, investment casting	Rough; large slow gears
Forming	Form milling cutter (one cutter per teeth no. range)	Moderate; job work
Generating	Hobbing, gear shaping, rack cutters - mass production, accurate involute	Good (most used)
Finishing	Shaving, grinding, lapping, honing, burnishing	High accuracy, quiet automotive gears
Others	Powder metallurgy (sintered), forging, plastic moulding (nylon), rolling	Light-duty gears, cheap

Example 4 - second design problem (with service factor)

> Design a spur gear pair for P = 10 kW, N1 = 720 rpm, i = 2.5, medium shock (Cs = 1.25), steel σb = 140 MPa, 20° FD, z1 = 20, z2 = 50, b = 10 m.

> Design torque $T = 9.55 \times 10^6 \times 10 \times 1.25/720 = 165{,}800$ N·mm. y = 0.1084. m^3 Cv = 2T/[z1 σb (b/m) π y] = 331,600/[20 × 140 × 10 × 0.1084 × π] = 34.8.

> m = 4 -> v = 3.02 m/s, Cv = 0.499, m^3 Cv = 31.9 < 34.8 ✗; m = 5 -> v = 3.77, Cv = 0.443, m^3 Cv = 55.4 ✓ -> **m = 5 mm**.

> d1 = 100, d2 = 250, a = 175 mm, b = 50 mm. Ft = 2T/d1 = 3316 N ; Fd = Ft/Cv = 7483 N ; Sb = 140 × 5 × 50 × 0.3405 = 11,918 N ✓.

> Wear: Q = 2 × 50/70 = 1.43 ; need K ≥ 7483/(b Q d1) = 7483/(50 × 1.43 × 100) = 1.05 -> 0.16 $(BHN/100)^2 \geq 1.05$ -> **BHN ≥ 256**. Choose C45 hardened & tempered, BHN 260 (Sw = 7726 N ✓).

Common mistakes to avoid in exam

> Using module in metres with forces in newtons - keep **mm and N** (stress in N/mm^2 = MPa).

> Forgetting the **service factor** when finding torque, or using Cv with the wrong velocity formula for the gear quality.

> Designing the stronger gear: always design the **weaker** (smaller σb × y).

> Not rounding module to the **standard value** and recomputing d, v, Cv and the checks.

> Checking only bending - wear (pitting) often controls for hardened pinions.

Efficiency, shaft design for gears

> Efficiency of a spur-gear pair (well lubricated) = **98-99%**; torque on gear T2 = i T1 η.

> Shaft: resultant bending moment from Ft (horizontal plane) & Fr (vertical plane): $M = \sqrt{(Mh^2 + Mv^2)}$; design with ASME code $d^3 = (16/\pi\ \tau)\ \sqrt{[(Kb\ M)^2 + (Kt\ T)^2]}$.

> Overhung pinions are avoided; keep bearing span so that deflection < 0.0005 rad at the gear for proper tooth contact.

> Gear blank: rim thickness ≥ 1.5-2 × tooth height; hub dia = 1.8 d_shaft; hub length = 1.2-1.5 d_shaft.

Modern (AGMA / ISO) approach in brief

> Bending stress: σ = Ft Ko Kv Ks Km / (b m J) - factors for overload, dynamics, size, load distribution; J = geometry factor.

> Contact stress: σc = Cp √(Ft Ko Kv Ks Km Cf / (b d1 I)) with Cp = 191 √MPa (steel pair); I = geometry factor for pitting.

> Allowable stresses corrected for life (cycles), reliability, temperature; safety factors 1.0–1.5 for bending and 1.0–1.2 for contact.

X. Quick revision & important questions

Formulas to remember

> $d = m\,z$; $p = \pi\,m$; $ha = m$; $hf = 1.25\,m$; $a = m\,(z1 + z2)/2$; $db = d\cos\varphi$; $z_{min} = 17$ ($20°$ FD).

> $T = 9.55 \times 10^6\,P/N$; $Ft = 2T/d$; $Fr = Ft\tan\varphi$; $Fn = Ft/\cos\varphi$; $v = \pi\,d\,N/60000$.

> $\sigma b = Ft/(b\,m\,Y)$, $Y = \pi\,y$; $y = 0.154 - 0.912/z$; $Sb = \sigma b\,m\,b\,Y$; $Fd = Ft/Cv$; $Cv = 3/(3 + v)$.

> $Sw = b\,Q\,d1\,K$; $Q = 2z2/(z1 + z2)$; $K = 0.16\,(BHN/100)^2$; Design: $Sb, Sw \geq Fd$.

> $m^3 = 2T/(z1\,\sigma b\,Cv\,(b/m)\,\pi\,y)$; $P = Ft\,v/1000$.

1. Explain gear terminology with a neat sketch and define module, pressure angle. (5 M)
2. State the law of gearing. What is interference? How can it be avoided? (5 M)
3. Derive Lewis equation for bending stress in spur gear tooth. (10 M)
4. Explain the dynamic load on gear tooth (Barth and Buckingham). (5 M)
5. Derive Buckingham wear load / Hertz contact stress for gears. (10 M)
6. How do you select a gear material based on bending and contact stresses? (5 M)
7. Design a spur gear pair for given power, speed and ratio. (10 M)
8. Determine the power transmitting capacity of a given spur gear pair. (10 M)